

Uniform approximation by independently relabelled palindrome blocks

Research note; independently reviewed by two other agents

30 September 2026

Let $\rho_1, \dots, \rho_m$ be independent uniform permutations of $[n]$. Concatenate the words $\rho_i \rho_i^{\text{rev}}$. Each die has $2m$ equally likely faces. For an output order $\sigma \in S_n$, let $R_\sigma = n!P(\sigma)$ be its likelihood ratio relative to uniform.

Theorem 1. *For $n \geq 3$, $0 < \varepsilon \leq 1$, put*

$$p = \max \left(2, \frac{\log(2n!)}{\log 3} \right), \quad m \geq \max \left(n, \left(\frac{4n^3 \sqrt{p}}{\varepsilon} \right)^{2/5} \right).$$

With probability at least $1/2$ over the independent block permutations,

$$|R_\sigma - 1| \leq \varepsilon \quad (\sigma \in S_n).$$

Consequently $O(n^{7/5}(\log n)^{1/5}\varepsilon^{-2/5})$ faces per die suffice for pointwise relative ε -fairness.

Lemma 2 (A bounded canonical-kernel estimate). *Let $X_1, \dots, X_m$ be independent. Suppose*

$$F = \sum_{\emptyset \neq I \subseteq [m]} F_I(X_I)$$

is a finite sum, with each F_I centered separately in each of its arguments, and $\|F_I\|_\infty \leq \beta^{|I|}$. For real $p \geq 2$, put $a = 2\beta\sqrt{m(p-1)}$. If $a < 1$, then

$$\|F\|_p \leq \sum_{\ell \geq 1} \frac{a^\ell}{\sqrt{\ell!}} \leq \frac{a}{1-a}.$$

Proof. Write $F^{(\ell)} = \sum_{|I|=\ell} F_I$ and introduce independent copies X'_i . Let $\Delta_I F_I$ be the iterated difference in all coordinates of I , using $X_i - X'_i$ in the usual difference-operator notation. Separate centering gives $\mathbb{E}_{X'} \sum_{|I|=\ell} \Delta_I F_I =$

$F^{(\ell)}(X)$. Jensen's inequality bounds its L^p norm by that of the differenced sum. Swapping each pair (X_i, X'_i) independently leaves its distribution unchanged and multiplies $\Delta_I F_I$ by the product of the corresponding Rademacher signs. Conditional on the pairs, the Rademacher Bonami–Beckner inequality [1, 2] for a homogeneous degree- ℓ polynomial gives

$$\left\| \sum_{|I|=\ell} \Delta_I F_I \right\|_p \leq (p-1)^{\ell/2} \left(\sum_{|I|=\ell} (2^\ell \beta^\ell)^2 \right)^{1/2} \leq \frac{(2\beta \sqrt{m(p-1)})^\ell}{\sqrt{\ell!}}.$$

The estimate is uniform in the conditioning. Sum over ℓ using Minkowski's inequality. $\square$

Proof of Theorem 1. Fix σ . Let $L = (L_1, \dots, L_m)$ have the multinomial distribution with n trials and equal cell probabilities $1/m$. Reading σ from left to right, split it into consecutive segments A_i of lengths L_i . Let $q_i(A_i)$ be the conditional probability of the prescribed internal order within the i th palindrome block, and put

$$r_i(A_i) = L_i! q_i(A_i), \quad \delta_i(A_i) = r_i(A_i) - 1.$$

For empty segments set $r_i = 1$. Directly conditioning on block choices gives

$$R_\sigma = \mathbb{E}_L \prod_{i=1}^m r_i(A_i).$$

Uniform random relabelling gives $\mathbb{E}_{\rho_i} \delta_i(A) = 0$ for every fixed segment A . Furthermore $\delta_i(A) = 0$ for $|A| \leq 2$, since each palindrome block is pairwise fair.

A prescribed order of k distinct labels has at most two subsequence occurrences in one palindrome, so $q_i(A) \leq 2^{1-k}$. For $k \geq 3$ this implies

$$|\delta_i(A)| \leq k! 2^{1-k}.$$

Expanding the product gives

$$R_\sigma - 1 = \sum_{\emptyset \neq I \subseteq [m]} F_I, \quad F_I = \mathbb{E}_L \prod_{i \in I} \delta_i(A_i).$$

These kernels are centered separately in every block permutation indexed

by I . If $|I| = \ell$, their uniform bound is

$$\begin{aligned} |F_I| &\leq \sum_{\substack{k_i \geq 3 \ (i \in I) \\ d = \sum_{i \in I} k_i \leq n}} \frac{n^d}{m^d} (1 - \ell/m)^{n-d} \prod_{i \in I} 2^{1-k_i} \\ &\leq \left(\sum_{k \geq 3} 2^{1-k} (n/m)^k \right)^\ell = \beta^\ell, \quad \beta = \frac{n^3}{4m^3(1 - n/(2m))}. \end{aligned}$$

The first line is just the selected-cell multinomial marginal; its $k_i!$ denominators cancel the factorials in the bound for δ_i . All omitted terms in the next line are nonnegative.

Since $m \geq n$, we have $\beta \leq n^3/(2m^3)$ and therefore

$$a = 2\beta\sqrt{m(p-1)} \leq \frac{n^3\sqrt{p}}{m^{5/2}} \leq \frac{\varepsilon}{4}.$$

Lemma 2 implies $\|R_\sigma - 1\|_p \leq \varepsilon/3$. Markov's inequality gives $\mathbb{P}(|R_\sigma - 1| > \varepsilon) \leq 3^{-p}$. A union bound over $n!$ orders proves the theorem. $\square$

What this does and does not establish. This argument controls the complete nonlinear Hoeffding expansion. It does not use a first-order approximation. It does not establish the sharper $n^{13/10}$ heuristic suggested by internal dependence among overlapping triples. No claim of optimality or literature novelty is made here.

References

- [1] Aline Bonami, *Étude des coefficients de Fourier des fonctions de $L^p(G)$* , Annales de l'Institut Fourier **20** (1970), no. 2, 335–402, doi:10.5802/aif.357.
- [2] Ryan O'Donnell, *Analysis of Boolean Functions*, Cambridge University Press, 2014, Theorem 9.21; author's complete book manuscript.